



SOFTWARE ENGINEERING

Class#6

CONTENT

- Systems Analysis and Design
- Software Engineering (Programming)
- Requirement Engineering
 - Activity
- Software Engineering Methodology
 - Waterfall Model
 - Agile
 - Scrum

WHY DO WE USE AN IT SYSTEM ?



in a business

SYSTEMS ANALYSIS & DESIGN

WHY IS IT IMPORTANT?

System

A collection of related **components** that interact to perform a task in order to accomplish a goal

Analysis & Design

Steps and procedures to ascertain how a system works and how make it better

SYSTEMS ANALYSIS & DESIGN

WHO ARE PARTICIPANTS IN THE PROJECT?

- **Users** – The system should always be developed in consultation with users. If user involvement is inadequate, the system may fail for lack of acceptance.
- **Management** – Managers should also be consulted.
- **Technical staff** – IS members (e.g. systems analysts, programmers) need to be involved, otherwise an outside IS people contracted to do the job.

Systems Analyst - an information specialist who performs systems analysis, design, and implementation.

- IS systems Analyst
- Study information and communication needs for an organization
 - Determine what changes are required to deliver better information



SYSTEMS DEVELOPMENT

THE 6 PHASES OF SYSTEMS ANALYSIS & DESIGN

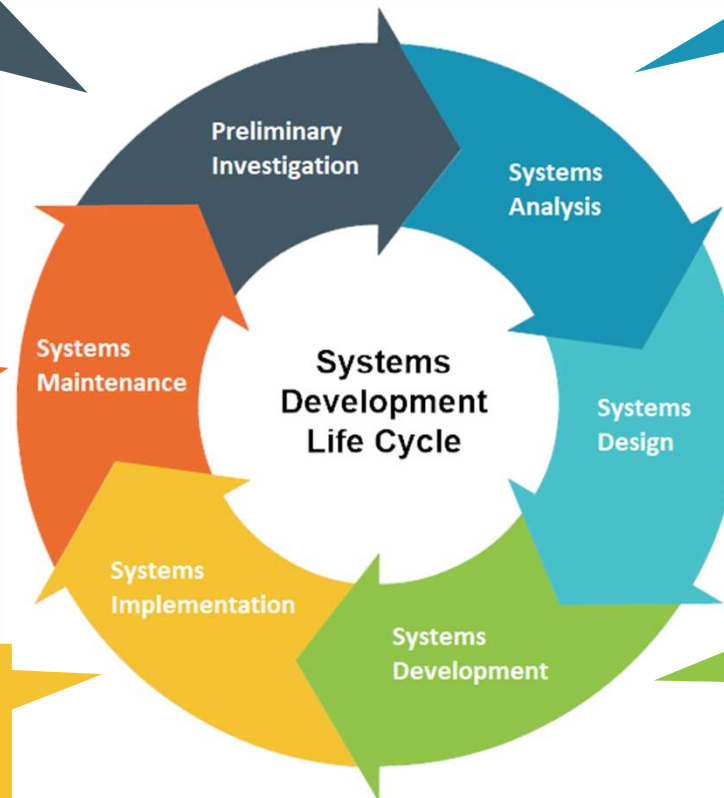
1. Conduct the preliminary analysis
2. Propose alternative solutions
3. Describe the costs & benefits
4. Submit a preliminary plan

The result may indicate it is the time to start the entire SDLC again.

1. Audit the system, adjust and improve the system
2. Evaluate the system periodically

- Direct, Parallel, Phased, and Pilot implementation

1. Convert to the new system
2. Train the users



Modeling Tools

-Analytical tools such as charts, tables, and diagram used to represent system
Ex. **Data Flow Diagram (DFD)**

- Requirements for new system

1. Gather data
2. Analyze the data
3. Write a report

- Major components and its' function
- S: - Output, input, storage, and processing requirements
- C
- P: - Systems control and backup

1. Do a preliminary design
2. Do a detail design
3. Write a report

- Make or buy

- Unit test or system test

1. Develop or acquire the software
2. Acquire hardware
3. Test the system

WHAT ARE COMMON REQUIREMENTS FOR A LOGIN AND SIGNUP PAGE ?



A login form titled "CS Tutor Login" with a light gray background and rounded corners. It contains two input fields: "User Name:" and "Password:". Below the fields is a "Login" button.

CS Tutor Login

User Name:

Password:

Login

WHAT ARE COMMON REQUIREMENTS FOR A LOGIN AND SIGNUP PAGE ?

facebook

Email

Password

☐ Keep me logged in

[Forgot your password?](#)

Facebook helps you connect and share with the people in your life.



Sign Up

It's free and always will be.

First Name:

Last Name:

Your Email:

Re-enter Email:

New Password:

I am:

Select Sex:

Birthday:

Month:

Day:

Year:

Why do I need to provide my birthday?

[Create a Page for a celebrity, band or business.](#)

English (US) Español Português (Brasil) Français (France) Deutsch Italiano العربية हिन्दी 中文(简体) 日本語 ...

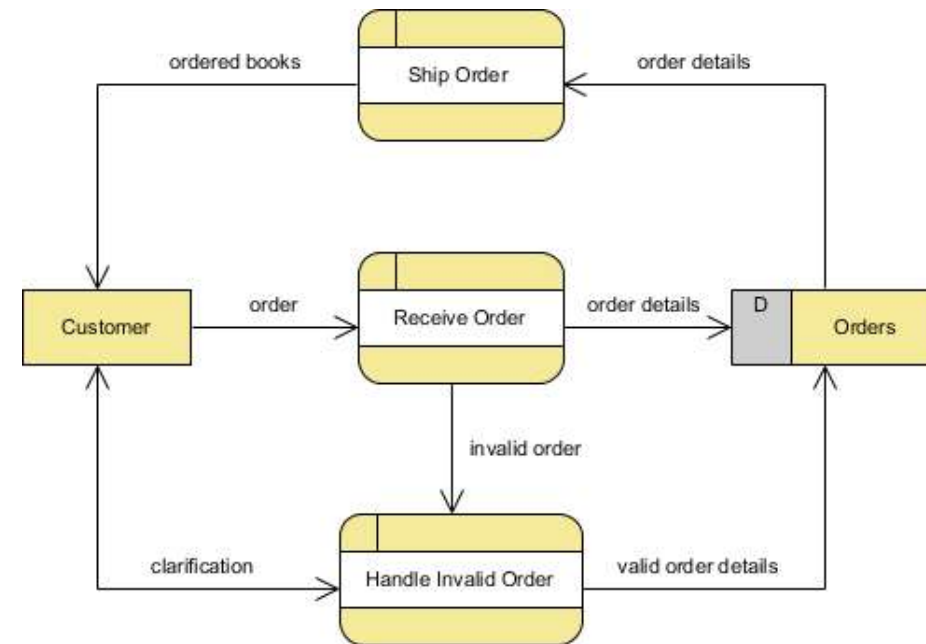
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DATA FLOW DIAGRAM - DFD

- structured analysis and design method
- widely used for software analysis and design
- depict a good amount of the system requirements graphically

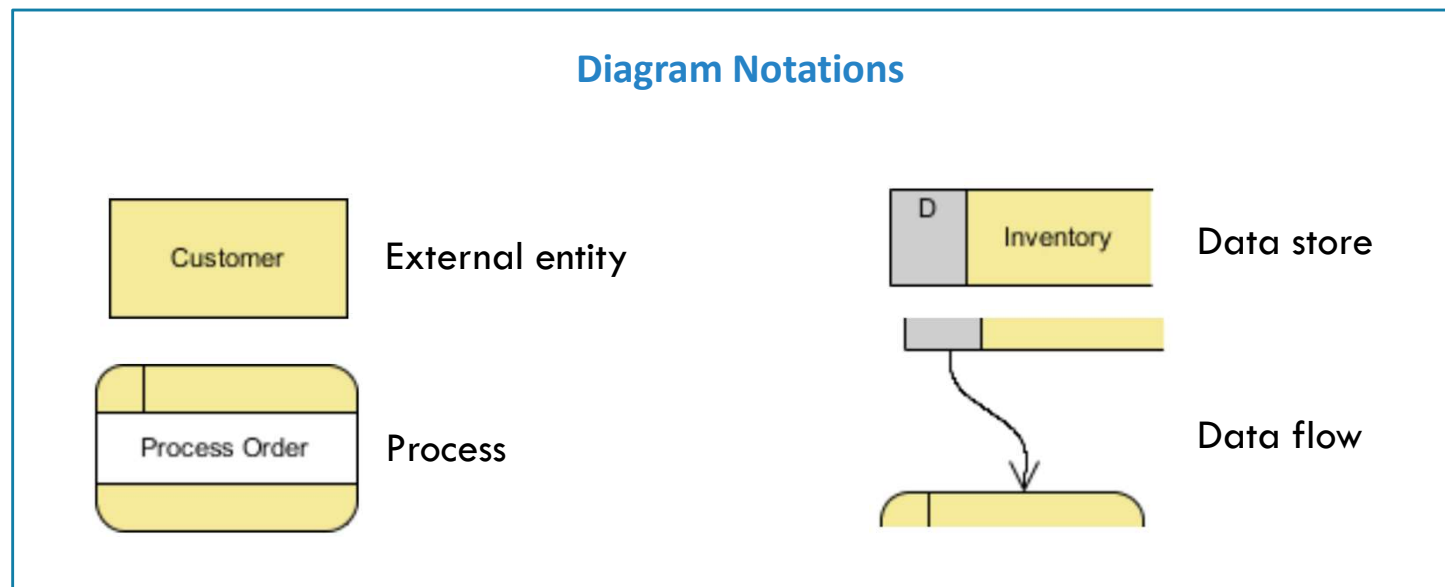
DFD includes following characteristics:

1. supporting the analysis and requirement stage of system design
2. a diagramming technique with annotation
3. describing a network of activities/processes of the target system
4. stepwise refinement through multi-level decomposition of processes



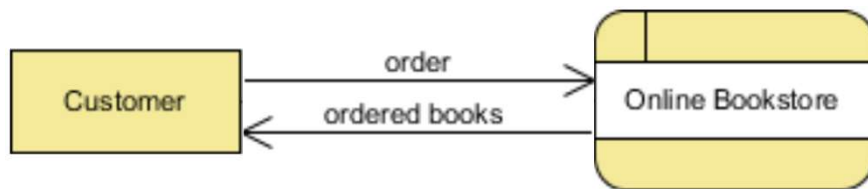
TAKE A LOOK AT DATA FLOW DIAGRAM

DFD presents a symbol system to describe [data flows](#) and a decomposition mechanism to describe a system at various detail levels.



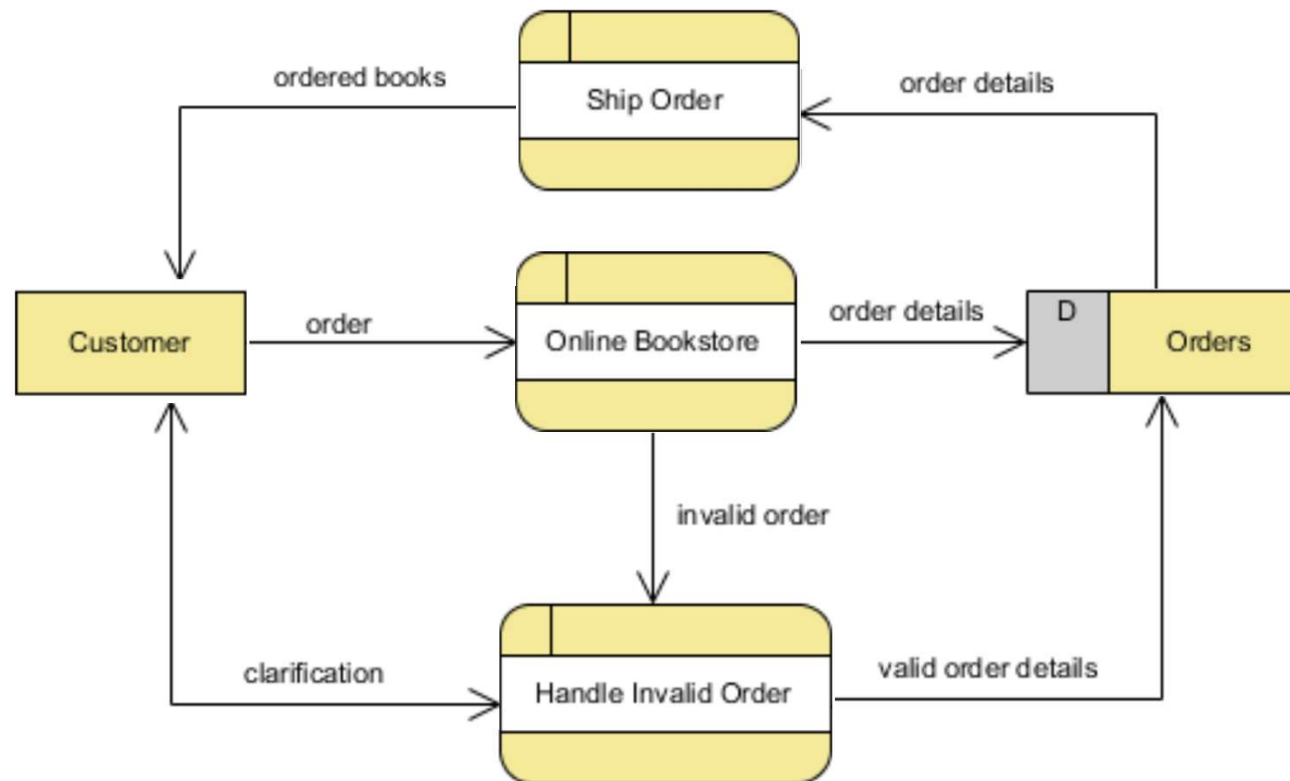
DFD EXAMPLE (ONLINE BOOKSTORE)

CONTEXT LEVEL (LEVEL 0) DFD



DFD EXAMPLE (ONLINE BOOKSTORE)

LEVEL 1 DFD



WHAT IS SOFTWARE ENGINEERING? [1]

- Includes the development and building of computer systems software and applications software
 - Computer systems software - programs that include computing utilities and operations systems
 - Applications software - user-focused programs that include web browsers, database programs, computer games, mobile applications, etc.
- Software engineers have extensive knowledge of programming languages, software development, and computer operating systems, and they apply engineering principles to software creation.

WHAT DOES A SOFTWARE ENGINEER DO ?

- analyse the client's needs
- design
- test; and
- develop the computer software in order to meet those needs

SOFT SKILLS MATTER IN SE ?

Key success factors

- able to write good documentation
- be a good communicator and able to voice your opinions in meetings
- possess some program management skills (Synergy)
 - getting people organized
 - trying to make them keep you up to date

SOFTWARE ENGINEERING (PROGRAMMING)

TRADITIONAL 5 STEPS OF SOFTWARE ENGINEERING

Step	Activities
Step 1: Problem clarification	1. Clarify program objectives and program users 2. Clarify desired outputs. 3. Clarify desired inputs. 4. Clarify desired processing. 5. Double-check feasibility of implementing the program. 6. Document the analysis.
Step 2: Program design	1. Determine program logic through topdown approach and modularization, using a hierarchy chart. 2. Design details using pseudocode and/or using flowcharts, preferably on the basis of control structures. 3. Test design with structured walkthrough.

SOFTWARE ENGINEERING (PROGRAMMING)

TRADITIONAL 5 STEPS OF SOFTWARE ENGINEERING

Step	Activities
Step 3: Program coding	1. Select the appropriate high-level programming language. 2. Code the program in that language, following the syntax carefully.
Step 4: Program testing	1. Desk-check the program to discover errors. 2. Run the program and debug it (alpha testing). 3. Run real-world data (beta testing).
Step 5: Program documentation and maintenance	1. Write user documentation. 2. Write operator documentation. 3. Write programmer documentation. 4. Maintain the program.



SUMMER MEADOW

4 minutes

Don't start until I say "GO"

EASTER EGGS

EASTER EGGS

Team setup:

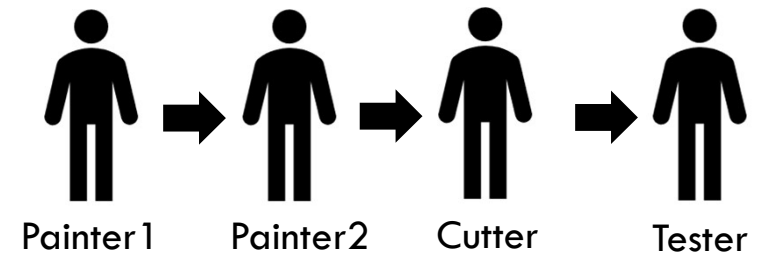
- Team size: 4 members: 1 cutter, 2 painters, 1 tester(QA)

Rules of the Game

1. Each egg must have at least two different colors
2. Two separate people must complete each coloring activity
3. Each egg should be minimum 90% filled with color
4. White space doesn't count as color
5. Cutting must be around oval edges of the egg
6. Eggs with major distractions in cutting will be disqualified

How to Win

The team that produces the greatest number of colorful eggs will win (with minimum waste).



Round 1: Every member will do their task in sequence

Task 1 (3 min): Plan who will do what, plan how many eggs will your team make e.g. X eggs

Task 2 (6 min):

- 1st painter paints X eggs according to your plan.
- 2nd painter continues painting after the 1st painter finished X eggs.
- The cutter cuts all finished eggs after the 2nd painter finished painting.
- The tester verifies quality of the finished eggs (drop disqualified eggs, only collect and turn in the good ones.)

Plan	Execution	
3 Min.	6 Min.	3 Min.

EASTER EGGS

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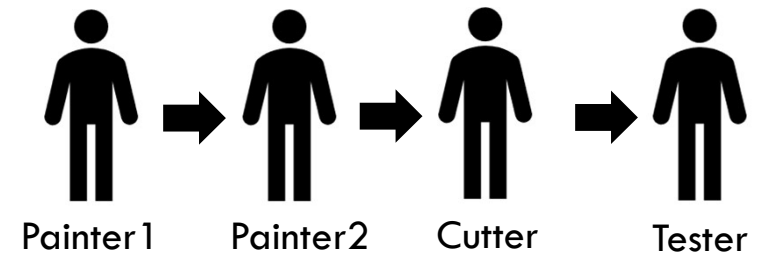
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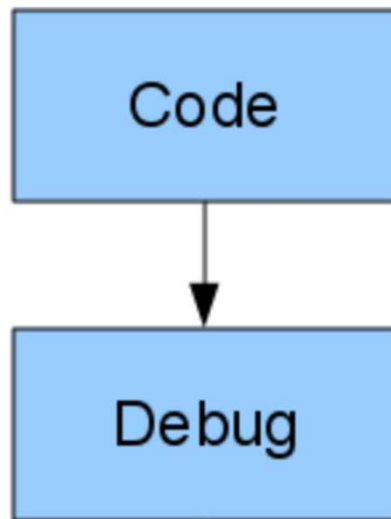
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SOFTWARE DEVELOPMENT METHODOLOGIES/ SOFTWARE PROCESS MODEL ^[2]

There are several software development methodologies/software process models

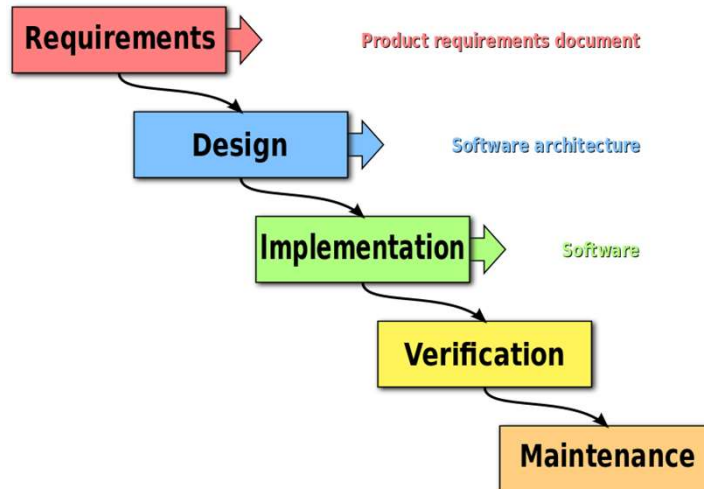
- Typical Student Programming Process



SOFTWARE DEVELOPMENT METHODOLOGIES/ SOFTWARE PROCESS MODEL ^[2]

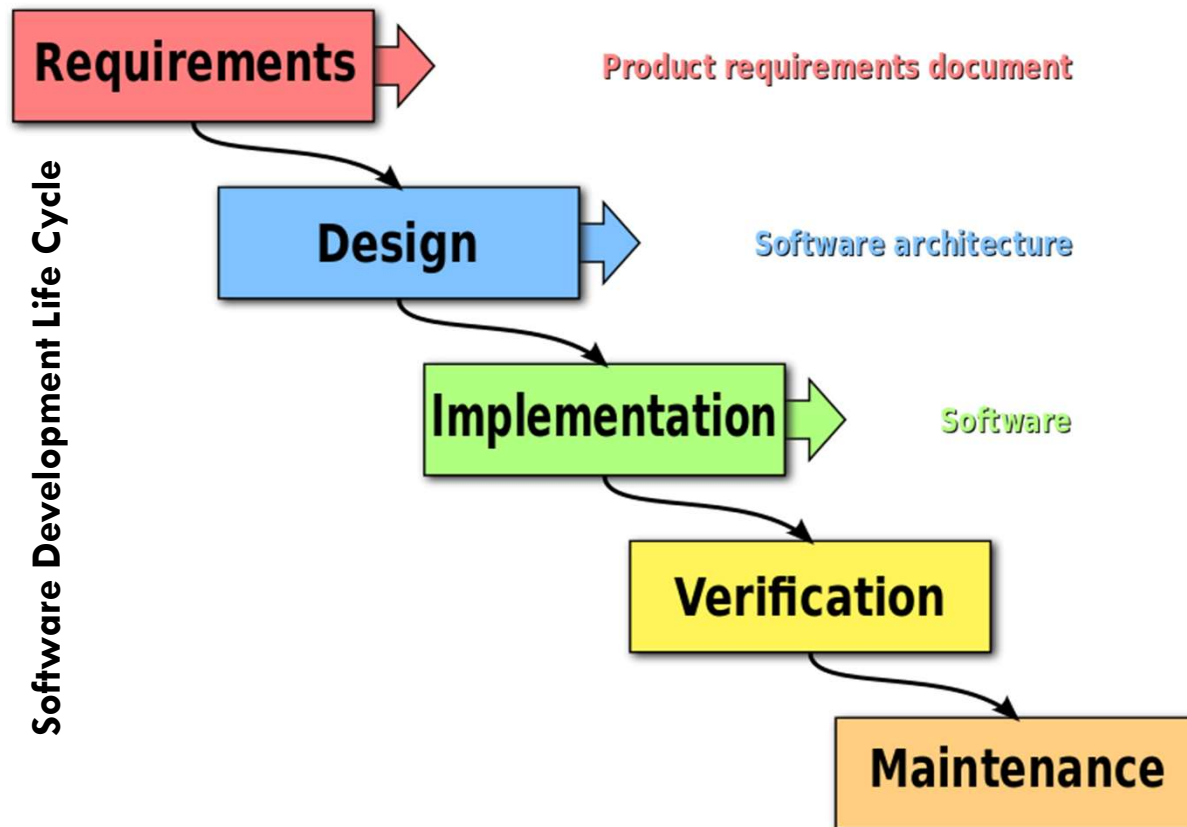
There are several software development methodologies/software process models

- Waterfall Model



SOFTWARE DEVELOPMENT METHODOLOGIES

WATERFALL MODEL is a sequential development approach, in which development is seen as flowing steadily downwards (like a waterfall) through several phases.



The waterfall model is more suitable for the highly structured physical environments meant that design changes became prohibitively expensive much sooner in the development process such as manufacturing and construction industries.

WATERFALL MODEL - APPLICATIONS

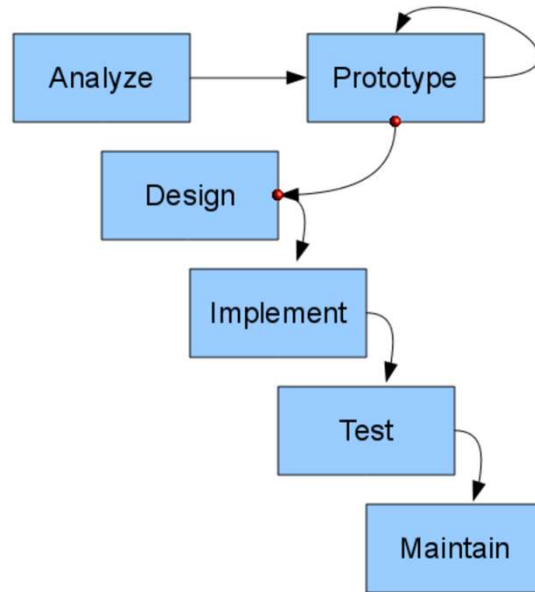
Waterfall model is most appropriate for

- Requirements are very well documented, clear and fixed.
- Product definition is stable.
- Technology is understood and is not dynamic.
- There are no ambiguous requirements.
- Ample resources with required expertise are available to support the product.
- The project is short.

SOFTWARE DEVELOPMENT METHODOLOGIES/ SOFTWARE PROCESS MODEL ^[2]

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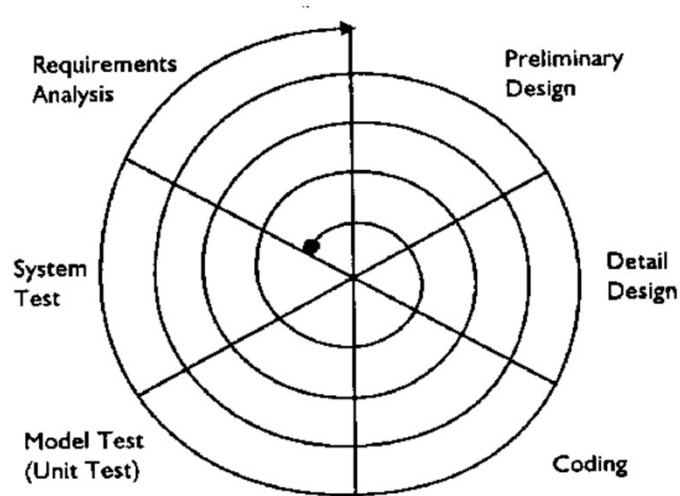
- Prototyping Model



SOFTWARE DEVELOPMENT METHODOLOGIES/ SOFTWARE PROCESS MODEL ^[2]

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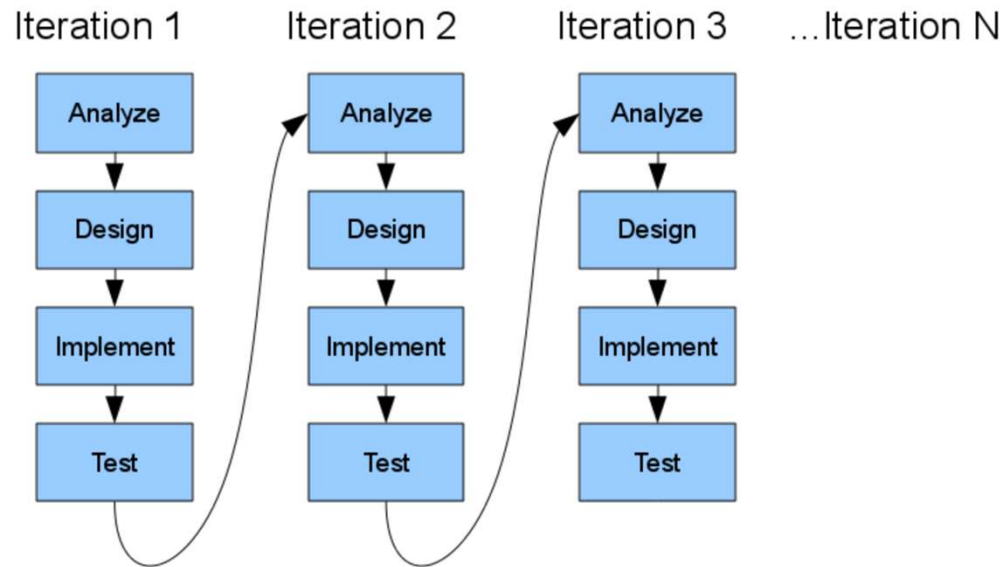
- Spiral Model



SOFTWARE DEVELOPMENT METHODOLOGIES/ SOFTWARE PROCESS MODEL ^[2]

There are several software development methodologies/software process models

- Agile Model



SOFTWARE DEVELOPMENT METHODOLOGIES/ SOFTWARE PROCESS MODEL

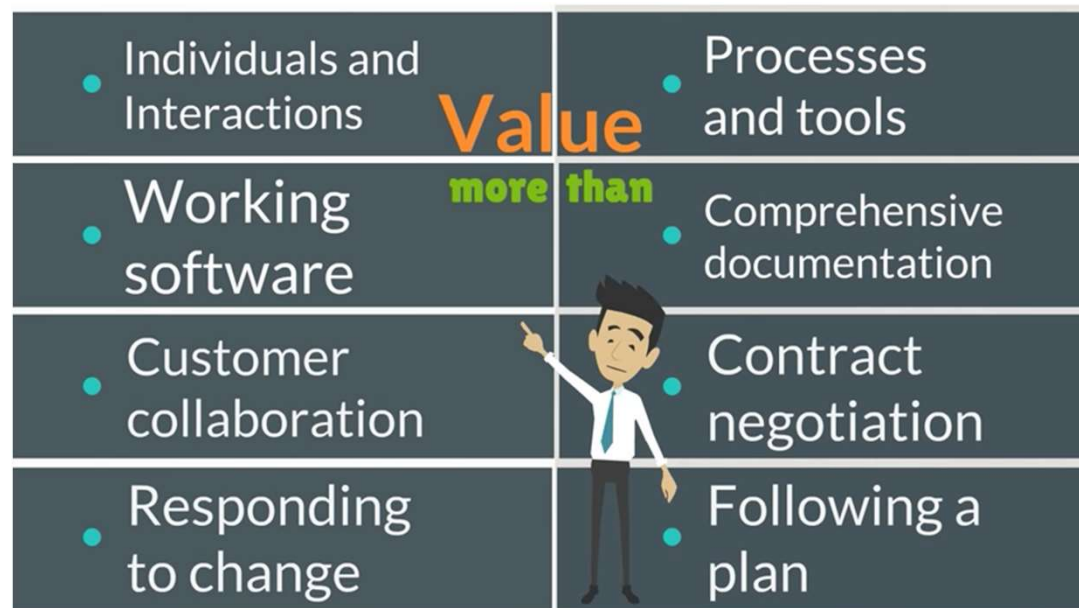
We discuss one of the basic method and the popular one here:

1. **Waterfall** might be more properly called the “traditional” approach, and
2. **Agile**, a specific type of Rapid Application Development and newer than Waterfall, but not that new, which is often implemented using Scrum.

SOFTWARE DEVELOPMENT METHODOLOGIES

AGILE is an iterative, team-based approach to development. This approach emphasizes the rapid delivery of an application in complete functional components

Agile Manifesto established a set of 4 values that are people-centric and result-driven.



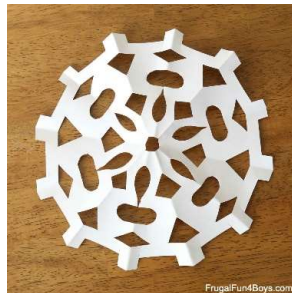
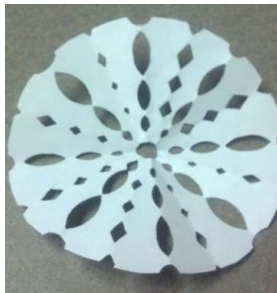


INTERMISSION

SOFTWARE DEVELOPMENT METHODOLOGY

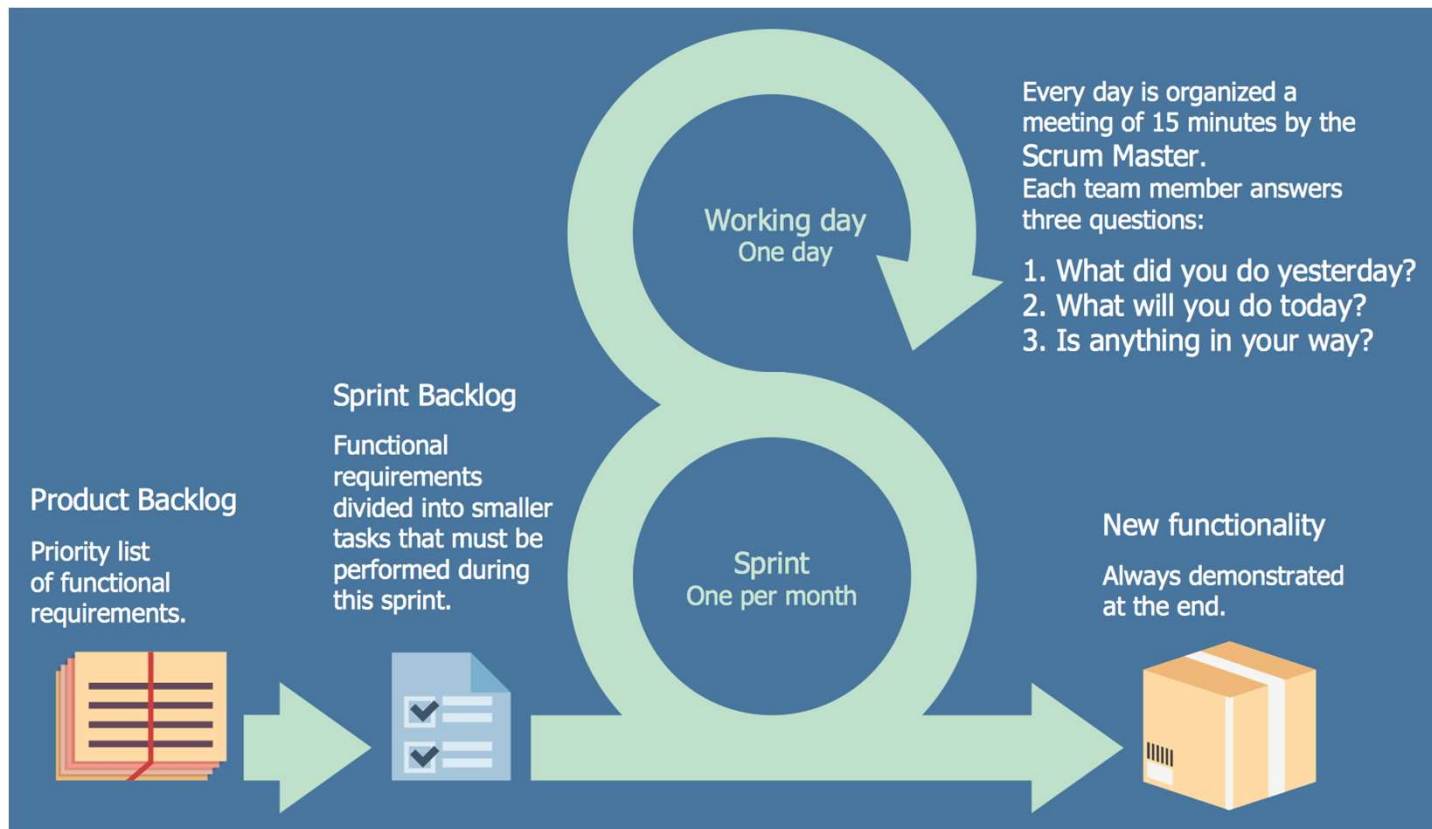
LEAN STARTUP [4]

- The **lean startup** methodology is used to develop products and businesses in a short period of time, which allows the creator of the product or business to quickly determine if their business model is a viable one.
- A startup that uses it will need to focus on obtaining customer feedback on the initial product. This feedback will help you make changes and iterations on the product that will allow it to be improved over time in accordance with the wishes of the customers.



SOFTWARE DEVELOPMENT METHODOLOGY

SCRUM is an agile implementation for managing a project (software development project) which depicts an iterative and incremental approach for the work on the complex projects.

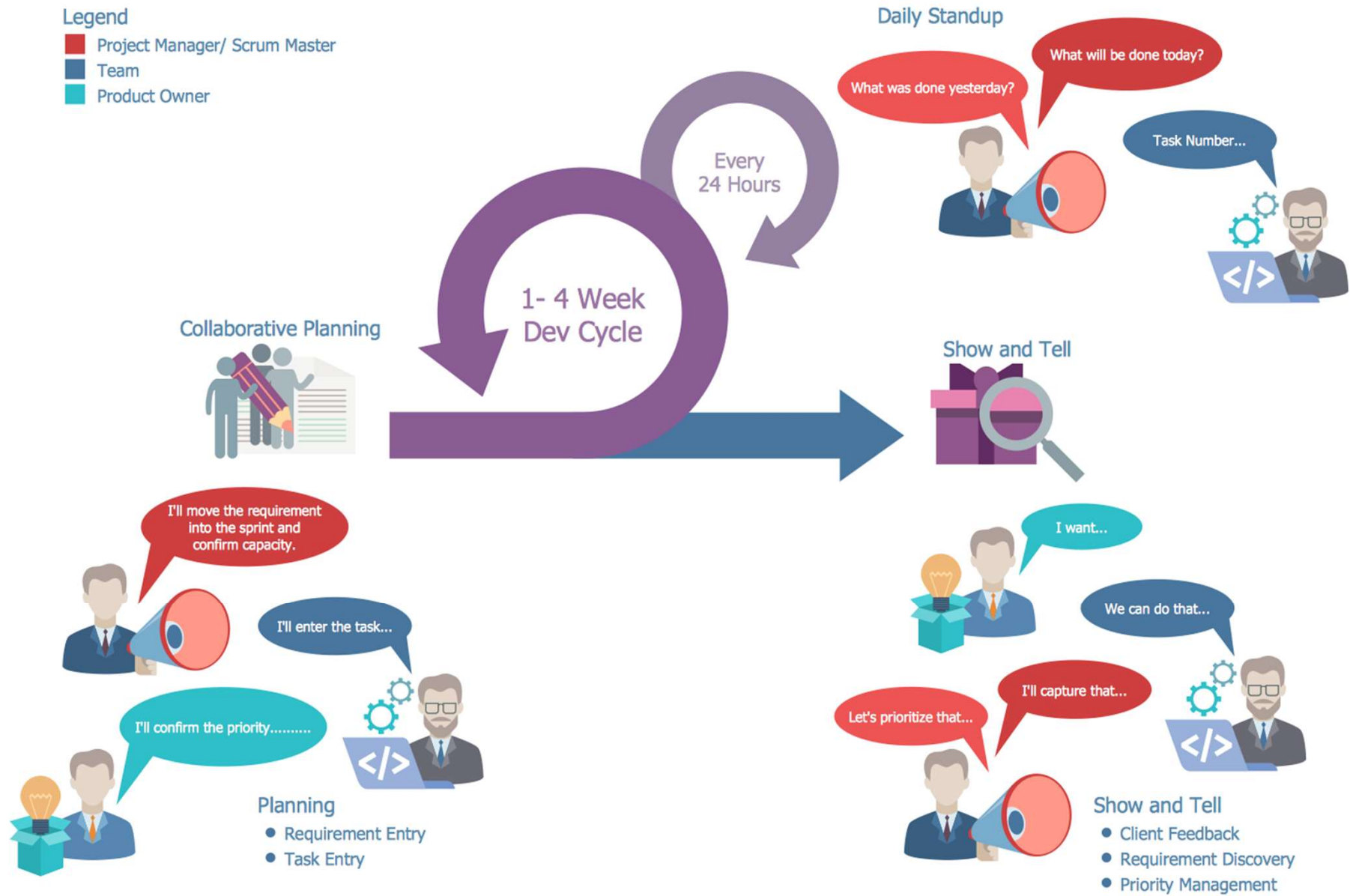


EXAMPLES OF COMPANIES USING SCRUM

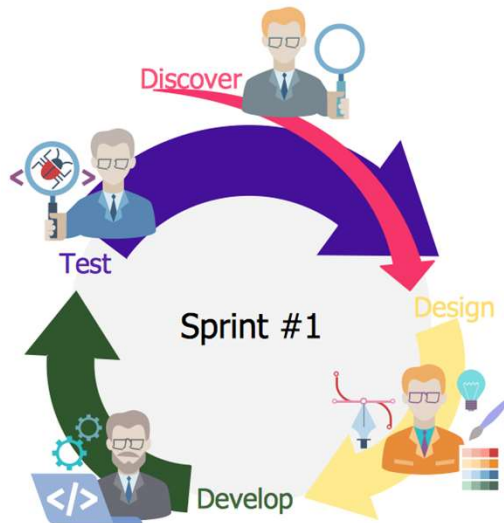


Legend

- Project Manager/ Scrum Master
- Team
- Product Owner



Agile methodologie



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[4] What Is the Lean Startup Methodology?,
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ADDITIONAL READINGS

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<https://www.360pmo.com/category/agile/>